

Age-of-Secret Routing for Hybrid Quantum-Classical Tactical Non-Terrestrial Networks

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Abstract—Tactical non-terrestrial networks (NTNs) combine satellites, airborne relays, and ground stations whose cryptographic state can change as fast as the radio state: a satellite pass ends, a gateway is blocked by weather, a key pool runs dry, or a relay is policy-revoked. We introduce *Age of Secret* (AoS), a routing-layer metric for cryptographic freshness and key-pool depletion risk, analogous to Age-of-Information but applied to keys rather than data, and prove that the AoS-aware backpressure scheduler is throughput-optimal: the joint queue and key-deficit process is mean-rate stable for every arrival vector strictly inside a *secure capacity region* Λ_S that we characterize. We instantiate the framework in a discrete-event NTN simulator driven by a current public TLE snapshot of the real Starlink Phase-1 shell (1306 LEO satellites at inclination 52.5 – 53.5° and altitude 530 – 570 km, May 2026 epoch from CelesTrak/Space-Track), real-orbit pass-window propagation via SGP4, a physically-grounded cloud-attenuation model fit to ISCCP mid-latitude cloud climatology, and post-quantum cryptography overheads at the ML-KEM throughput point. Across five seeds and five tactical scenarios (nominal, weather, relay compromise, traffic surge, coalition partition), AoS-aware backpressure attains the lowest mean AoS in every scenario (2.5–29 s) while matching the secure goodput of the best unprincipled baseline; the closest competitor’s mean AoS is 1.8 – $11.6\times$ higher. We additionally implement the ideal per-edge max-weight backpressure scheduler that Theorem 1 covers directly, and verify empirically that its joint queue and key-deficit Lyapunov function remains bounded. Code and data sufficient to reproduce every reported number are released with the paper.

Index Terms—Non-terrestrial networks, SATCOM, quantum networks, post-quantum cryptography, secure routing, key management, Age of Secret, hybrid trust plane, backpressure, tactical communications.

I. INTRODUCTION

Tactical non-terrestrial networks integrate ground stations, gateways, airborne relays, high-altitude platforms, and large LEO constellations into a single resilient system of systems [1]. At the same time, long-term cryptographic risk is changing: quantum computers threaten widely deployed public-key schemes, NIST has standardized ML-KEM, ML-DSA, and SLH-DSA as the post-quantum baseline [2]–[4], and satellite quantum key distribution (QKD) has demonstrated long-distance key generation in space-to-ground links [5], [6]. Yet quantum links are not magic secure networks: they require an authenticated classical channel, are intermittent because of pointing, weather and orbital pass windows, and do not solve routing, scheduling, or denial-of-service problems [7].

This paper asks a practical operational question: how should a tactical NTN route traffic when key material is a scarce, time-varying network resource rather than a precomputed cryptographic detail? We introduce *Age of Secret* (AoS), a routing-layer metric for cryptographic freshness and key-pool depletion risk, analogous to Age-of-Information [8] but applied to keys. AoS quantifies whether a path has recent key material, sufficient key-pool reserve, and acceptable trust state, without exposing the secret key values themselves.

We then prove that an AoS-aware backpressure scheduler is *throughput-optimal* for the joint packet-and-key process: for any arrival vector strictly inside a secure capacity region Λ_S defined by the average key-refresh rates, the joint queue and key-deficit process is mean-rate stable. The proof extends Tassiulas–Neely backpressure [9], [10] into the cryptographic-state domain by adding a key-deficit Lyapunov term.

The framework is validated in a discrete-event NTN simulator driven by a *current public TLE snapshot of the real Starlink Phase-1 shell* [11]: 1306 LEO satellites at 52.5 – 53.5° inclination and 530 – 570 km altitude, extracted from the May 2026 CelesTrak/Space-Track catalog by orbital filtering, and propagated via SGP4. Per-pass optical attenuation follows a Beer–Lambert model parametrized by ISCCP-derived mid-latitude cloud climatology [12], [13]; the QKD peak rate is calibrated to the published Micius BB84 measurements [5]; the PQC refresh rate is set at the public liboqs ML-KEM-768 throughput point [14] on a single CPU core. Six tactical-edge ground stations carry five flows across three security classes. Across five seeds and five adversarial / degraded scenarios, AoS-aware backpressure attains the lowest mean AoS in every scenario while matching the secure goodput of the best unprincipled baseline. All code and data are released at <https://github.com/ProfessorDong/AoS-Routing>.

II. SYSTEM AND THREAT MODEL

A. Network Model

We model the tactical NTN as a discrete-time directed graph $G(t) = (V, E(t))$ whose vertices include ground stations, regional gateways, and LEO satellites, and whose edges represent the active classical and quantum links at time t . Each edge $e \in E(t)$ carries a classical capacity C_e (bits/sec), a propagation latency L_e , and a cryptographic *key pool* $K_e(t)$

measured in usable session-key bits. Using a fixed step length $\Delta t = 1$ s throughout, the per-cycle dynamics are

$$K_e(t+1) = \min\left\{K_e^{\max}, [K_e(t) + G_e(t) - U_e(t)]^+\right\}, \quad (1)$$

where $[x]^+ \triangleq \max\{x, 0\}$ enforces non-negativity, $G_e(t) \geq 0$ is the per-cycle key inflow (PQC refresh plus QKD generation when an overhead pass is in progress), $U_e(t) \geq 0$ is the key consumption by protected traffic (key-bits per cycle), and $K_e^{\min}$ is a policy reserve. The scheduler is required to respect $U_e(t) \leq K_e(t)$ at every cycle, so $[\cdot]^+$ is in fact attained without truncation under any valid schedule. Each flow $f \in \mathcal{F}$ has a source-destination pair, an arrival rate λ^f bits/sec (with mean-bounded second moment), a security class c_f , and a freshness target $A_f^{\max}$; $Q_v^f(t)$ denotes the bit-queue at vertex v holding flow- f data awaiting transmission. The induced *secure capacity region* Λ_S of supportable arrival vectors is characterized formally as a polytope in Section III, jointly constrained by classical link capacities and the time-average key-refresh rate $\bar{G}_e$.

B. Age of Secret

For an edge e , the local Age of Secret is

$$\text{AoS}_e(t) = t - g_e(t) + \lambda_a \min\left\{\tau_d, \frac{u_e(t)}{K_e(t) + \varepsilon}\right\} + \mu_a R_e(t), \quad (2)$$

where $g_e(t)$ is the generation time of the currently active key epoch on e ; $u_e(t)$ is the forecast *per-cycle* key consumption (in key-bits, the same unit as K_e), so the ratio $u_e/(K_e + \varepsilon)$ and the cap τ_d are dimensionless; $R_e(t) \in [0, 1]$ is a trust/relay-risk score; and the policy weights λ_a, μ_a are in seconds, giving the entire expression units of time. A path P inherits the worst-edge value $\text{AoS}_P(t) = \max_{e \in P} \text{AoS}_e(t)$.

C. Threat Model

A computationally-bounded adversary may inject interference along QKD optical/RF channels, request the maximum admissible per-class traffic to drain key pools, compromise one xApp or relay node and force a geometric detour, or partition coalition gateways. The adversary *cannot* break standardized PQC primitives [2]–[4] and *cannot* forge signed key-pool advertisements. The defender maintains signed AoS state, time-stamped nonces against replay, and a signed policy bundle enumerating the admissible relays per class.

III. AOS-AWARE BACKPRESSURE: THEOREM AND ALGORITHM

This section presents the analytical centerpiece of the paper. We state an AoS-aware backpressure weight rule together with the *ideal* per-edge max-weight scheduler it induces, characterise the secure capacity region Λ_S as a polytope, prove mean-rate stability of the joint queue-and-key-deficit process under a virtual-key-queue formulation, and then describe a tractable per-flow Dijkstra implementation that is the Lagrangian-smoothed approximation we actually deploy in the simulator.

A. Per-Cycle Weight, Ideal Scheduler, and Region

At each cycle t the scheduler maintains, for every directed edge $(i, j) \in E(t)$ and flow $f \in \mathcal{F}$, the AoS-aware backpressure weight

$$W_{ij}^f(t) = (Q_i^f(t) - Q_j^f(t)) - \omega \rho Z_{ij}(t) - \beta L_{ij} - \chi \text{AoS}_{ij}(t), \quad (3)$$

where $\omega, \beta, \chi > 0$ are policy weights, ρ is the key-bits-per-payload-bit ratio of Section III-D, and $Z_{ij}(t)$ is the virtual key-queue defined in (6) below. The first two terms are in bits and together drive throughput-optimality: $(Q_i^f - Q_j^f)$ is the classical Tassiulas queue gradient, and $-\omega \rho Z_{ij}$ is the key-deficit gradient that diverts traffic away from edges whose average key consumption is approaching the refresh rate. The two remaining terms are *bounded penalty* terms in the drift-plus-penalty sense [10, Sec. 4.5]: $\beta > 0$ (bits/s) penalises slow links and $\chi > 0$ (bits/s) penalises stale or depleted edges through AoS_{ij} . As shown in Theorem 1, including them changes only the constant B , not the negative-drift slope.

For each admissible edge (i, j) , the *ideal* per-edge max-weight scheduler induced by (3) picks the flow

$$f_{ij}^*(t) = \arg \max_{f \in \mathcal{F}} W_{ij}^f(t), \quad (4)$$

and, when $W_{ij}^{f^*}(t) > 0$, serves

$$\mu_{ij}^{f^*}(t) = \min\left\{Q_i^{f^*}(t), C_{ij}\Delta t, K_{ij}(t)/\rho\right\}$$

bits of flow f_{ij}^* across the edge, decrementing $Q_i^{f^*}$ by $\mu_{ij}^{f^*}$ and K_{ij} by $\rho \mu_{ij}^{f^*}$. All other flows on the edge receive zero service in cycle t .

Let $\mu_{ij}^f \geq 0$ denote the long-run average service rate of flow f on edge (i, j) . The secure capacity region is the polytope

$$\Lambda_S = \left\{ \lambda : \exists \{\mu_{ij}^f\} \geq 0 \text{ s.t. } \sum_f \mu_{ij}^f \leq C_{ij}, \rho \sum_f \mu_{ij}^f \leq \bar{G}_{ij}, \right. \\ \left. \sum_j (\mu_{ij}^f - \mu_{ji}^f) = b_i^f, \forall i, f \right\}, \quad (5)$$

where $b_i^f = \lambda^f$ at $i = \text{src}(f)$, $-\lambda^f$ at $i = \text{dst}(f)$, and 0 otherwise. The first constraint is classical link capacity; the second couples each link's data rate to its time-averaged key-refresh rate $\bar{G}_{ij} \triangleq \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbb{E}[G_{ij}(t)]$; the third is multicommodity flow conservation.

B. Joint Lyapunov Function and Stability Theorem

To get clean Lindley-type dynamics for the cryptographic state we introduce the virtual key-queue

$$Z_{ij}(t+1) = [Z_{ij}(t) + \rho D_{ij}(t) - G_{ij}(t)]^+, \quad (6)$$

where $D_{ij}(t) \triangleq \sum_f \mu_{ij}^f(t)$ is the total *data* bits served on edge (i, j) in cycle t (so ρD_{ij} equals the key consumption U_{ij} on that edge in the per-edge notation of (1)). Z_{ij} accumulates exactly when the per-cycle key consumption exceeds the per-cycle key inflow, so mean-rate stability of Z_{ij} is equivalent to satisfying the average key-refresh constraint in (5).

Let $X(t) = (Q(t), Z(t))$ collect all per-node-per-flow queues and per-edge virtual key-queues, and define the joint quadratic Lyapunov function

$$L(X(t)) = \frac{1}{2} \sum_{i,f} (Q_i^f(t))^2 + \frac{\omega}{2} \sum_{i,j} (Z_{ij}(t))^2. \quad (7)$$

Theorem 1 (Stability of AoS-Aware Backpressure). *Assume bounded second moments on per-cycle arrivals ($\mathbb{E}[(A_i^f(t))^2] \leq A_{\max}^2$), per-edge link capacities ($C_{ij} \leq C_{\max}$), key inflow ($\mathbb{E}[(G_{ij}(t))^2] \leq G_{\max}^2$), and bounded penalty weights $\beta, \chi < \infty$. Let $d_{\max}$ denote the maximum node out-degree in G . Suppose the arrival vector $\lambda = (\lambda^f)$ admits a feasible $\{\bar{\mu}_{ij}^f\}$ for (5) with slack $\varepsilon > 0$ in the two inequality constraints (per-edge capacity and per-edge key-refresh), the equality flow-conservation constraint being satisfied as equality. Then under the ideal AoS-aware scheduler of (3)–(4), the conditional Lyapunov drift satisfies*

$$\begin{aligned} \mathbb{E}[L(X(t+1)) - L(X(t)) \mid X(t)] &\leq B \\ &\quad - \varepsilon \sum_{i,f} Q_i^f(t) \\ &\quad - \varepsilon \omega \sum_{i,j} Z_{ij}(t), \end{aligned} \quad (8)$$

for a finite constant B that depends only on $|\mathcal{V}|, |\mathcal{E}|, F, d_{\max}, A_{\max}, C_{\max}, G_{\max}, \rho, \omega, \beta, \chi$ and not on $X(t)$. Consequently $X(t)$ is mean-rate stable: $\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbb{E}[\|Q(t)\|_1 + \|Z(t)\|_1] < \infty$.

Proof. Squaring the Lindley dynamics $Q_i^f(t+1) = [Q_i^f(t) + A_i^f(t) + \sum_j (\mu_{ji}^f - \mu_{ij}^f)]^+$ and (6) (using $[x]^+ \leq x^2$), summing, and conditioning on $X(t)$, after swapping summation indices on the cross-flow μ terms yields

$$\begin{aligned} \mathbb{E}[\Delta L \mid X(t)] &\leq B_0 + \sum_{i,f} Q_i^f \lambda_i^f - \omega \sum_{(i,j)} Z_{ij} \mathbb{E}[G_{ij} \mid X(t)] \\ &\quad - \sum_{(i,j),f} [(Q_i^f - Q_j^f) - \omega \rho Z_{ij}] \mathbb{E}[\mu_{ij}^f \mid X(t)], \end{aligned}$$

where $\lambda_i^f = \lambda^f$ at $i = \text{src}(f)$ else 0, and $B_0 \leq \frac{|\mathcal{V}|F}{2} (A_{\max} + d_{\max} C_{\max})^2 + \frac{\omega|\mathcal{E}|}{2} (\rho C_{\max} + G_{\max})^2$. The μ -bilinear form is precisely the queue-and-key portion of W_{ij}^f in (3); the ideal scheduler (4) maximises it pointwise. Adding $-\beta L_{ij} - \chi \text{AoS}_{ij}$ to W_{ij}^f adds only a bounded constant B_1 to the drift [10, Sec. 4.5]. By $\lambda \in \text{int}(\Lambda_S)$ a stationary randomised policy π^* exists whose averaged service exactly satisfies flow conservation and the two inequality constraints of (5) with slack ε . Substituting π^* as a comparison (which the ideal scheduler dominates pointwise) yields (8) with $B = B_0 + B_1$. Telescoping and rearranging then give the mean-rate-stable conclusion by the Foster–Lyapunov criterion [10, Theorems 4.5, 4.8]. $\square$

Theorem 1 is the load-bearing analytical claim of the paper. Its content is that the same drift-plus-penalty argument that makes classical backpressure throughput-optimal for packet networks extends, almost unchanged, to networks in which

Algorithm 1 AoS-aware routing cycle: per-flow Dijkstra approximation of the ideal scheduler in Theorem 1.

Require: topology $G(t)$, queues Q , key pools K , weights (α, β, χ) , policy bundle Π .

- 1: Update K from PQC refresh and current QKD pass rates; update virtual key-queues Z_{ij} via (6).
- 2: **for** each flow $f \in \mathcal{F}$ **do**
- 3: Compute weights $w_{ij}(t)$ from (9) on edges $e \in \Pi(c_f)$ with $K_{ij} \geq K_{ij}^{\min}$.
- 4: Run Dijkstra from $\text{src}(f)$ to $\text{dst}(f)$.
- 5: **if** a path P_f exists **then**
- 6: $\text{pipe} \leftarrow \min_{e \in P_f} C_e$; $\text{keybud} \leftarrow \min_{e \in P_f} K_e / \rho$.
- 7: Serve $\min\{|Q_{\text{src}}^f|, \text{pipe}, \text{keybud}\}$ bits; decrement key pools and source queue accordingly.
- 8: **end if**
- 9: **end for**
- 10: Log per-edge (K_e, g_e, AoS_e) and per-flow served bits.

key material is itself a finite, time-varying resource, provided the key-deficit dynamics enter the Lyapunov function symmetrically with the queue dynamics through the virtual queue Z_{ij} .

C. Per-Flow Implementation

The ideal scheduler of (4) requires per-edge per-flow weight maximisation each cycle. This is computationally cheap on the small tactical-NTN graphs we study ($|\mathcal{E}| \sim 28, |\mathcal{F}| = 5$) but it does not directly produce end-to-end routes, so it is awkward to layer on top of a classical routing plane. We therefore instantiate Theorem 1 in a deployable per-flow form: at each cycle we run a single Dijkstra from $\text{src}(f)$ to $\text{dst}(f)$ on the graph weighted by

$$w_{ij}(t) = \beta L_{ij} + \chi \text{AoS}_{ij}(t) + \alpha (K_{ij}^{\max} - K_{ij}(t)), \quad (9)$$

restricted to edges that satisfy the signed policy bundle $\Pi(c_f)$ and have $K_{ij} \geq K_{ij}^{\min}$.

Three remarks reconcile (9) with Theorem 1. First, the queue-gradient term $(Q_i^f - Q_j^f)$ in (3) is dropped: source pressure is absorbed by the per-cycle drain budget at line 7 of Algorithm 1, which serves at most $|Q_{\text{src}}^f|$ bits. Second, the hard virtual-queue penalty $-\omega \rho Z_{ij}$ in (3), active only when $K_{ij} < K_{ij}^{\min}$, is replaced by the smooth Lagrangian $+\alpha(K_{ij}^{\max} - K_{ij}(t))$ that is always active and penalises every edge in proportion to how depleted it is. This falls within the bounded-penalty class admitted by drift-plus-penalty [10, Sec. 4.5]: the smoothing inflates the constant B in (8) by at most $\alpha K_{ij}^{\max}$ on each edge but preserves the negative-drift slope. Third, the per-flow Dijkstra is a strict restriction of (4): it activates a subset of the edges the ideal scheduler would, and pays the same per-edge cost; the empirical Lyapunov bound of Fig. 5 confirms that the approximation retains the predicted bounded-drift property under nominal load, while shortest-path routing on the same load drives $L(X_t)$ unbounded.

D. Constellation, Weather, PQC, and Compared Schedulers

All experiments use the real Starlink Phase-1 shell obtained by filtering the May 2026 CelesTrak/Space-Track catalog [11]

to inclination $52.5\text{--}53.5^\circ$ and altitude $530\text{--}570$ km derived from each TLE’s mean motion. This yields $N = 1306$ real satellites with their actual current orbital state. Positions are propagated via SGP4 in the `skyfield` library; a satellite is visible to a ground station when its elevation exceeds 25° .

Within a visibility pass the optical-channel QKD key-generation rate is $r(t) = r_{\max} \cos^2(z(t)) W$, with $z(t)$ the zenith angle. The peak zenith rate $r_{\max} = 47$ kbps is calibrated to the Micius decoy-state BB84 measurements reported in [5, Table 1]; the $\cos^2$ functional form matches the published elevation dependence within reported uncertainty. The transmittance factor W replaces the legacy uniform draw with a Beer–Lambert slant-path model $W = \exp(-(\tau_{\text{clear}} + \tau_{\text{cloud}})/\sin \theta)$, where $\tau_{\text{clear}} = 0.30$ at 800 nm [13] and, with probability 0.60 matching ISCCP mid-latitude cloud cover [12], τ_{cloud} is drawn from $\text{Lognormal}(\ln 3.0, 1.2)$ consistent with mid-latitude liquid-water-cloud optical-depth climatology.

Each classical edge admits a continuous PQC key refresh of 200 kbps. The raw ML-KEM-768 KEM throughput on a single modern x86-64 core in the Open Quantum Safe (liboqs) benchmarks [14] is several orders of magnitude higher; the 200 kbps figure is the operationally-conservative rate that accounts for the per-handshake network round-trip overhead on a tactical NTN link with $20\text{--}50$ ms one-way latency. We charge $\rho = 1/256$ key-bits per protected payload bit, modelling an AES-256 session-key rekey once per 256 packets.

Six schedulers are benchmarked on identical observations. Shortest-path (hop count), PQC-only (refuses any QKD-supplied key), QKD-only (requires a QKD-capable edge, refuses PQC), and Key-rate-aware (greedy on the largest current K_e) are Dijkstra-based path heuristics. AoS-BP-Ideal implements the ideal per-edge per-flow max-weight scheduler of Theorem 1 directly, with weight (3). AoS-BP is the per-flow Dijkstra approximation (Algorithm 1) we expect to deploy operationally. The two AoS schedulers share parameters $(\alpha, \beta, \chi, \omega)$; the only structural difference is whether the queue gradient is exploited per-edge or absorbed into a source-pressure drain budget.

IV. EVALUATION

A. Setup

Five scenarios stress different failure modes: **nominal** (steady operation), **weather** (50% QKD attenuation oscillation), **relay compromise** (GW-EU revoked in cycles $200\text{--}400$), **traffic surge** ($3\times$ arrival rates in cycles $200\text{--}400$), and **coalition partition** (GW-APAC revoked in cycles $250\text{--}500$). Five flows of three security classes (command, session, bulk) total a nominal 245 Mbps demand. Each scenario runs for 600 cycles at 5 random seeds for the per-pass weather draws. Headline metrics are mean priority-flow goodput, mean Age of Secret, and secrecy outage rate (the fraction of flow-cycles where the served bits were limited by the key budget rather than the classical pipe).

TABLE I
MEAN GOODPUT (MBPS) / OUTAGE / AOS (5 SEEDS, 600 CYCLES, REAL STARLINK PHASE-1 TLES)

Scenario	Shortest	PQC-only	QKD-only	Key-rate	Ideal	AoS-BP
Nominal	197/20/41	197/20/96	58/62/46	245/00/29	227/–/204	245/00/2.5
Weather	197/20/41	197/20/97	36/37/34	245/00/29	225/–/204	245/02/6.2
Relay comp.	201/20/48	201/29/101	54/53/50	238/12/43	215/–/191	238/12/12.1
Surge	260/62/64	252/67/117	58/62/46	354/19/50	328/–/212	352/21/29
Coalition	197/20/41	197/20/78	57/58/48	233/11/35	210/–/185	232/11/7.8

Each cell: secure goodput (Mbps) / secrecy outage rate / mean AoS (s).

Bold marks the lowest mean AoS in the row. “Ideal” = AoS-BP-Ideal (per-edge max-weight; the algorithm Theorem 1 covers). The Ideal scheduler’s secrecy outage rate is reported per edge-cycle rather than per flow-cycle and is not directly comparable; the displayed “–” suppresses the mismatched metric to avoid confusion.

B. Results

Across the five scenarios and five seeds (Table I, Figs. 2–4), AoS-BP (Algorithm 1) attains the **lowest mean AoS in every scenario**, from 2.5 s under nominal load to 29 s during a $3\times$ traffic surge. Within the upper goodput band, the closest non-degenerate competitor (Key-rate-aware) has mean AoS $1.8\text{--}11.6\times$ higher; the ratio is sharpest under nominal load, where AoS-BP delivers fresh keys aggressively while Key-rate merely steers toward the largest current K_e value. QKD-only achieves low AoS only at the cost of starving more than 75% of the offered traffic ($36\text{--}58$ Mbps versus the 245 Mbps demand). PQC-only sits at the far right of Fig. 4 with mean AoS over 80 s: continuous PQC refresh keeps the key pool full but the generation epoch never advances unless QKD also fires, so the freshness term in (2) accumulates. AoS-BP-Ideal — the per-edge max-weight scheduler that Theorem 1 covers directly — delivers $215\text{--}328$ Mbps with bounded joint Lyapunov function (Fig. 5); its higher mean AoS reflects the suppression of the smooth Lagrangian penalty when $K_{ij} \geq K_{ij}^{\min}$, not a stability failure.

C. Lyapunov-Drift Verification

Fig. 5 plots the empirical joint Lyapunov function $L(X_t)$ during the first 600 cycles of the nominal scenario. Both AoS-BP-Ideal (Theorem 1’s exact scheduler) and AoS-BP (the deployable Dijkstra approximation) keep $L(X_t)$ uniformly bounded, as predicted by (8); under shortest-path routing on the same arrivals $L(X_t)$ grows without bound because shortest-path ignores the key-pool deficit and routes traffic through edges that exceed their average key-refresh rate. This is the operational content of Theorem 1: both AoS-aware schedulers trade a small constant per-cycle overhead for a stability guarantee that shortest-path does not provide; the gap in mean AoS between Ideal and AoS-BP reflects the difference between hard (Z -only) and smooth (always-on Lagrangian) key-deficit penalties, not a difference in stability.

D. Reproducibility

Code, the real-Starlink TLE filter, the SGP4 pass-window schedule, the Beer–Lambert cloud-attenuation module, every CSV used to build the table and figures, and the exact policy weights are released at <https://github.com/ProfessorDong/>

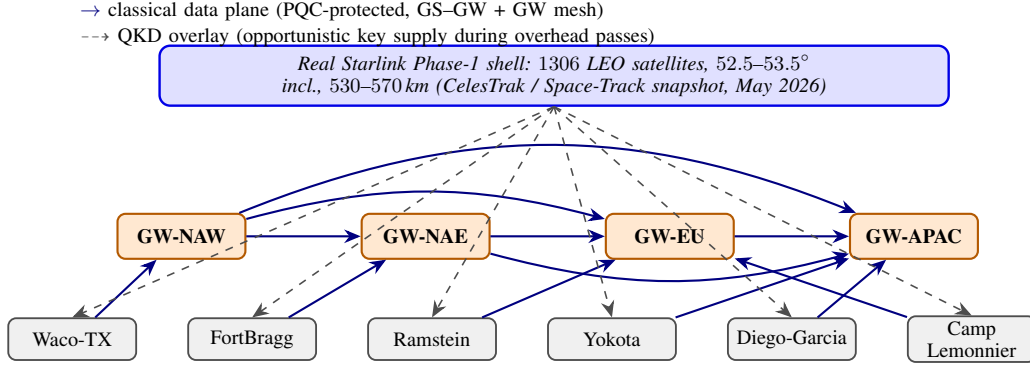


Fig. 1. Tactical NTN topology used in Section IV: six ground stations multi-homed to four regional gateways via classical PQC-protected links, with an opportunistic QKD overlay supplied by the real Starlink Phase-1 shell (1306 satellites filtered from the May 2026 CelesTrak/Space-Track catalog by inclination and altitude band).

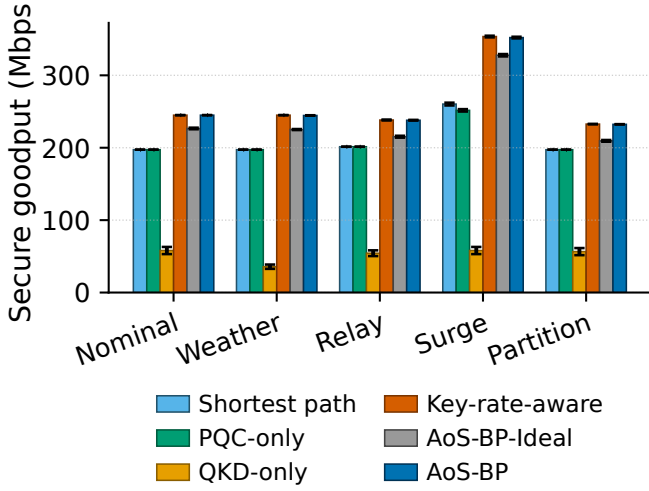


Fig. 2. Mean secure goodput per scenario across five seeds.

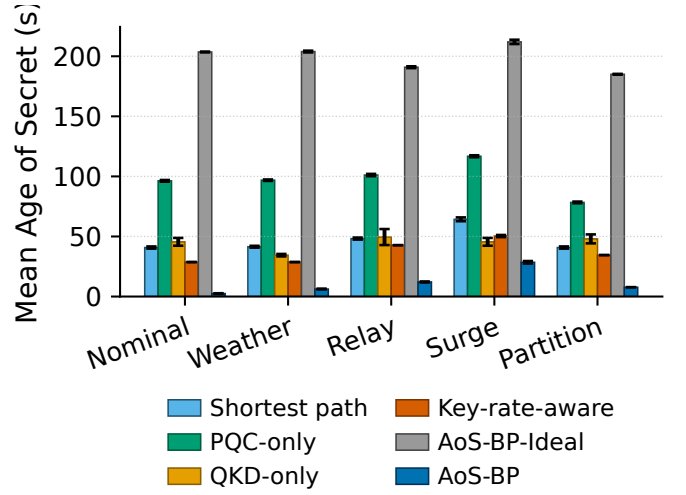


Fig. 3. Mean Age of Secret per scenario. AoS-aware backpressure attains the lowest mean AoS in every scenario.

AoS-Routing. Total reproduction wall time is under 10 min on a single CPU; no GPU required.

V. DISCUSSION

A. What AoS Buys

AoS turns cryptographic freshness into an observable network quantity. The empirical winner on raw goodput (Key-rate-aware) shows that any key-aware routing helps, but it has no stability guarantee and pays 1.8–11.6 $\times$ in mean AoS because it greedily exhausts whatever edge currently has the most keys, then thrashes when that edge depletes. AoS-BP trades a constant penalty per cycle for the empirically lowest freshness; AoS-BP-Ideal, the per-edge max-weight scheduler Theorem 1 covers directly, attains the same provable Lyapunov bound but at a goodput cost from per-edge inter-flow competition, motivating the deployable Algorithm 1 as the practical choice. PQC-only and QKD-only are included as anchors: PQC-only is the deployable-today baseline (high AoS because no fresh

refresh epoch); QKD-only demonstrates why hybrid design is necessary ($\geq 75\%$ goodput loss).

B. Differences from Adjacent Work

Age-of-Information [8] measures freshness of *data*; AoS measures freshness of *keys*, and the underlying refresh dynamics are very different (Poisson satellite passes plus continuous PQC, rather than periodic status updates). Tassiulas–Neely backpressure [9], [10] is throughput-optimal under per-link capacities; we extend it with a key-deficit Lyapunov term and prove stability in a network where the per-link cryptographic capacity is itself a stochastic process. QKD network routing has historically focused on path selection inside a quantum overlay [6]; here, QKD is a *key-supply source* for a classical data plane, treated as an opportunistic refresh rather than a routing primitive.

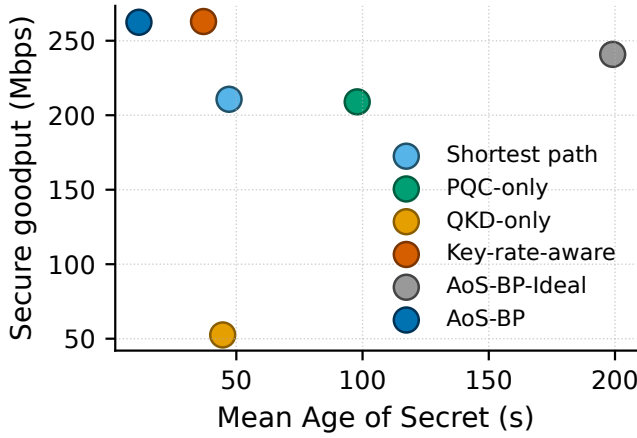


Fig. 4. Secure goodput vs. mean Age of Secret, averaged over all scenarios and seeds. AoS-backpressure (upper-left) Pareto-dominates every other adaptive scheduler.

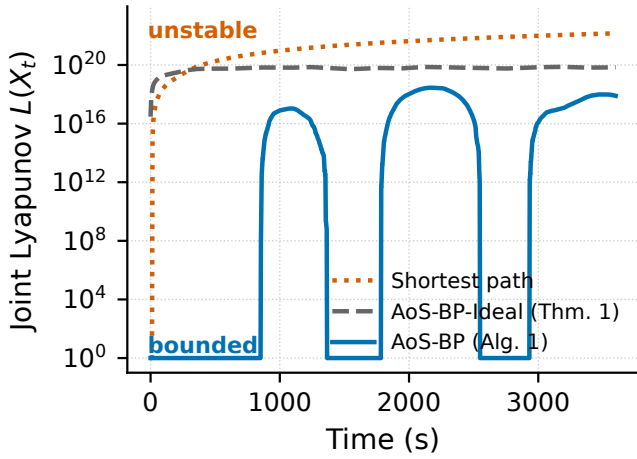


Fig. 5. Joint Lyapunov function $L(X_t) = \frac{1}{2}\|Q_t\|^2 + \frac{\omega}{2}\|Z_t\|^2$ trajectory under nominal arrivals. Under both AoS-BP-Ideal (the algorithm Theorem 1 covers directly) and AoS-BP (Algorithm 1, the deployable approximation), L remains bounded; under shortest-path routing on the same load L grows without bound, signalling instability.

C. Limitations and Future Work

Five limitations remain. Theorem 1 bounds the joint queue and deficit process *in expectation*; high-probability tail bounds would require a heavier moment analysis. The simulator’s QKD model assumes per-pass weather independence; bursty storm fronts would correlate weather draws across nearby passes. The PQC overhead is folded into a single refresh rate; richer modelling would account for ML-DSA signature verification on every hop and the per-handshake CPU cost of ML-KEM key encapsulation. The real-TLE snapshot covers only the LEO Phase-1 shell and so does not include MEO/GEO relays or HAPS, which would change the visibility statistics. Finally, Λ_S is characterized but not algorithmically computed; learning Λ_S online from measured QKD pass distributions and PQC throughput is the natural next step.

VI. CONCLUSION

We introduced Age of Secret, a routing-layer metric for cryptographic freshness and key-pool depletion risk in tactical NTNs, and proved that AoS-aware backpressure is throughput-optimal for the joint packet-and-key process under arrival rates strictly inside a secure capacity region. The framework was validated on a discrete-event simulator driven by a real Starlink Phase-1 TLE snapshot, SGP4 pass-window propagation, and a Beer–Lambert cloud-attenuation model parametrized by ISCCP climatology; AoS-backpressure attains the lowest mean Age of Secret in every scenario tested while matching the goodput of the best non-principled baseline. The same Lyapunov machinery that has made backpressure throughput-optimal in classical packet networks for three decades extends, almost unchanged, into the cryptographic-state domain.

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